

To drive down the cost per watt of solar energy, it is important to achieve high flexibility and overall equipment effectiveness. Equally important is to effectively and rapidly adopt new recipes and cell structures being developed in R&D labs

Breakage rate is part of typical TCO calculation in semiconductor industry, but only as a cost of the wafers themselves. However, broken wafers can incur other costs that directly affect profits. Every broken wafer is missed potential revenue from a finished cell, plus the lost production time for cleaning up the broken silicon pieces. This adds up to lost profit.

Fig. 2 models annual loss due to

TABLE I
Total Annual Losses Due to Wafer Breakage for a Line with a 0.3% Breakage Rate vs a Line with a 0.1% Breakage Rate

(Assuming the same number of wafers produced in both cases and value of wafer \$2.10)

	0.3% Breakage rate	0.1% Breakage rate
Wafers produced over one year without breakage (millions)	~24	~24
Breakage rate	0.3%	0.1%
Average line stop time (seconds)	15	5
Number of broken wafers	72,000	24,000
Yearly loss of production, number of wafers	864,000	96,000
Net loss due to broken wafers over one year (\$)	149,300	49,770
Net loss due to lack of profit (\$)	366,450	40,720
Total loss (\$)	517,750	90,490
Yearly benefit (\$)		425,260

wafer breakage, taking breakage rate and equipment downtime into consideration.

Table I provides a picture of the true costs of broken wafers. A fab that can lower its breakage rate by 0.2 percent can see a potential profit increase of more than \$0.5 million per year.

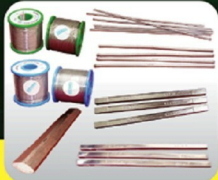
Print quality. Print quality is also an important factor in overall solar cell yield. Proper print alignment is increasingly important as the industry moves toward denser interconnects and techniques such as selective emitter and double printing. Cells with improperly aligned printing must often



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